

Table of Contents

01	Executive Summary
02	Problem Statement
03	Introducing Kratos
04	System Architecture
05	Multi-Agent Architecture
06	RAG Pipeline & Knowledge Base
07	Technology Stack
08	IBM Technology Integration
09	Real-Time Match Advisor Workflow
10	Visualization Dashboard
11	Kratos Conversational Assistant
12	Backend Architecture
13	Frontend Architecture
14	Security & Non-Functional Requirements
15	Project Deliverables Checklist
16	Future Scope & Limitations
17	Conclusion

01 Executive Summary

Kratos is a multi-agent AI system that reads sports data the way a coaching staff does — and turns it into a matchday plan. Built for **IBM Hackathon Problem Statement No. 29**, it pairs Retrieval-Augmented Generation over a dedicated football knowledge base with four specialist agents, coordinated by a single orchestrator agent named **Kratos**, running on IBM watsonx Orchestrate and reasoning with IBM Granite-4.0-8B-Instruct on watsonx.ai.

Coaches interact with Kratos through one voice — a web dashboard, a chat assistant, or a REST API — while, behind the scenes, dedicated agents analyse past matches, profile the next opponent, recommend a tactical plan, and explain that plan in plain coaching language. This report documents the problem, the architecture, the agent design, the IBM technology mapping, and the full deliverable set for submission.

4

SPECIALIST AGENTS + 1 ORCHESTRATOR

1,000

HISTORICAL MATCHES INDEXED

100

TEAMS IN THE KNOWLEDGE BASE

300

PLAYERS TRACKED

02 Problem Statement

PROBLEM STATEMENT NO. 29 — AI GAME STRATEGY ADVISOR AGENT

Build a complete AI-powered multi-agent game strategy advisor agent that analyses historical sports matches, opponent strategies, player performance, and tactical formations using IBM AI technologies.

The solution must satisfy the full IBM Hackathon requirement set — from retrieval-augmented knowledge to a production-style multi-agent backend, a coach-facing visualization dashboard, and a conversational assistant that a real coaching staff could use before, during, and after a match.

Mandatory Technology Stack

IBM Orchestrate · IBM LangFlow · IBM watsonx.ai · IBM Granite-4.0-8B-Instruct · Python · FastAPI · ReactJS · Tailwind CSS · FAISS / ChromaDB · LangChain · Docker · IBM Cloud

Kratos maps every one of these requirements to a specific layer of the architecture described in this report — see Section 07 for the full technology mapping.

03 Introducing Kratos

In Greek, *kratos* means strength — the personification of raw, decisive power. It's a fitting name for the one agent identity every coach on the team will actually talk to: direct, data-backed, and unwilling to hedge when a decision is needed before kickoff.

Meet Kratos

Role — AI Game Strategy Advisor

Home platform — IBM watsonx Orchestrate

Reasoning engine — Granite-4.0-8B-Instruct (watsonx.ai)

Speaks through — Dashboard, Chat, REST API

Personality — Direct and tactically confident; always states which agent and which data a recommendation is built on.

Kratos is not a single model call — it's the front door to four specialist agents. When a coach asks a question, Kratos decides which specialist (or specialists) to consult, blends their findings, and answers in one consistent voice. The rest of this report describes that system end to end.

04 System Architecture

Kratos is built in five layers, from raw data to the surfaces coaches actually use.

DATA SOURCES Historical Matches · Player Statistics · Match Reports Tactical Formations · Coaching Notes · Tournament History RETRIEVAL-AUGMENTED GENERATION · FAISS VECTOR STORE Every record is embedded and indexed for real-time semantic retrieval by all agents Match Analysis – Possession & passing – Heatmaps & shot maps – Expected Goals (xG) – Set pieces & corners – Player ratings – Momentum trends Opponent Strategy – Formations & pressing – Counter & wing attacks – Strengths & weaknesses – Set-piece patterns – Key danger players Strategy Recommendation – Formation & starting XI – Attack & defense plan – Pressing intensity – Set-piece tactics – Risk & alternatives Insight Explanation – Why this formation – Why these players – Risks & strengths – Score probability – Match confidence KRATOS Orchestrator Agent IBM watsonx Orchestrate · delegates to all four specialist agents Dashboard ReactJS · Tailwind · Dark Mode Kratos Chatbot RAG-grounded conversational agent REST API FastAPI · JWT · Swagger

Figure 1 — Kratos system architecture. Data flows from the knowledge base through retrieval and the four specialist agents; Kratos, the orchestrator, synthesises their output and delivers it through the dashboard, chatbot, and API.

The Five Layers

- **Data Sources** — the raw football knowledge base: matches, players, reports, formations, notes, tournaments.
- **Retrieval Layer** — FAISS-backed RAG over embedded records, queried in real time.
- **Specialist Agents** — four independent agents, each with one narrow, well-defined job.
- **Kratos Orchestrator** — the IBM watsonx Orchestrate agent that delegates and synthesises.
- **Delivery Channels** — the dashboard, the chatbot, and the REST API.

05 Multi-Agent Architecture

Each agent has one job and does it well. Kratos calls on them individually or in combination, depending on the question being asked.

Agent A

Match Analysis Agent

Turns raw match data into a clear performance picture.

- Possession, passing accuracy & ball recovery
- Heatmaps, shot maps & Expected Goals (xG)
- Corners & set-piece analysis
- Winning trends & momentum shifts
- Player ratings across the squad
- Auto-generated match summaries

Agent B

Opponent Strategy Agent

Profiles how the next opponent plays — and predicts what they'll try.

- Formations & pressing style
- Counter-attacks, wing play & long balls
- Passing patterns & buildup habits
- Strengths, weaknesses & danger zones
- Set-piece strategy
- Most dangerous players, flagged by name
- Predicted opponent behaviour

Agent C

Strategy Recommendation Agent

Converts analysis into an actual game plan.

- Best formation & starting XI
- Defensive setup & attacking strategy
- Pressing intensity & player marking
- Substitution & counter-attack plans
- Corner, free-kick & penalty-order tactics
- Risk analysis with alternative strategies

Agent D

Insight Explanation Agent

Translates every AI decision into language a coaching staff trusts.

- Why this formation, why these players
- Why this tactic, in plain terms
- Expected risks & expected strengths
- Score probability & match confidence
- The final recommendation, explained

All four agents report to Kratos, the orchestrator — see Section 08 for how that agent itself is configured in IBM watsonx Orchestrate.

06RAG Pipeline & Knowledge Base

Every recommendation Kratos makes is grounded in retrieved evidence, not free-floating generation. The knowledge base spans seven record types, embedded and indexed in FAISS for real-time semantic search.

- Historical matches
- Player statistics
- Opponent strategies

- Match reports
- Tactical formations
- Coaching notes

- Tournament history

Knowledge Base at a Glance

Record type	Scale
Historical matches	1,000
Teams	100
Players	300
Player statistics, match reports & coach notes	Linked per match & player
Formation data & tournament data	Linked per team & competition
Opponent tactics	Linked per opponent team

At query time, the relevant agent embeds the question, retrieves the top matching records from FAISS, and passes them to Granite-4.0-8B-Instruct as grounding context alongside the agent's own prompt template — the same retrieval pattern LangChain provides out of the box, authored visually in LangFlow (see Section 08).

07Technology Stack

Every mandatory technology from the problem statement is mapped to a specific layer of Kratos.

Layer	Technology	Purpose
AI Reasoning	IBM watsonx.ai + Granite-4.0-8B-Instruct	Core language model behind every agent
Agent Orchestration	IBM watsonx Orchestrate	Hosts the Kratos agent: profile, delegation, channels
Visual Agent Design	IBM LangFlow	No-code pipeline for chat, RAG, agents, memory, retriever
Orchestration Framework	LangChain	Retrieval chains, prompt templates, agent tooling
Vector Store	FAISS (ChromaDB-compatible)	Embedding storage & semantic search
Backend	Python + FastAPI	REST APIs, authentication, business logic
Frontend	ReactJS + Tailwind CSS	Coach-facing dashboard
Containerisation	Docker	Reproducible local & cloud deployment
Cloud Platform	IBM Cloud	Hosting, scaling, managed services
Authentication	JWT	Stateless, token-based API security

08 IBM Technology Integration

IBM watsonx.ai & Granite-4.0-8B-Instruct

Granite-4.0-8B-Instruct, served through watsonx.ai, is the reasoning engine behind every agent in the system. Each of the four specialists and Kratos itself uses its own prompt template against the same model, and the same watsonx.ai environment provides the embedding model that feeds the FAISS vector store.

IBM watsonx Orchestrate — Kratos's Home

Kratos is configured as a complete agent inside IBM watsonx Orchestrate, ready to import and deploy:

Configuration field	What it defines
Profile	Name (Kratos), description, and avatar for the AI Game Strategy Advisor
Knowledge	Connected to the FAISS-backed football knowledge base
Behavior	Direct, evidence-first; every answer names its supporting agent and confidence
Instructions	System prompt governing when to delegate vs. answer directly
Tools	The four specialist agents, exposed as callable skills
Delegation	Routes a question to one or more agents and merges their responses
Channels	Web dashboard, embedded chat widget, and Slack/Teams-ready
Guidelines & Knowledge Source	Grounding rules and the linked knowledge base connection

Kratos — Welcome Message & Quick Prompts

Welcome message — “Hi, I'm Kratos, your AI Game Strategy Advisor. Ask me about an opponent, a formation, or what to change at half-time.”

Quick prompts — “How should we play against [Opponent]?” · “What's our best formation?” · “Where are we weakest right now?” · “Why this tactic?” · “Predict the scoreline.”

IBM LangFlow — Visual Pipeline

The RAG + agent + memory + retriever + output pipeline is authored visually in LangFlow: Chat → RAG → Agents → Memory → Retriever → Output. The flow exports as JSON, so it is versioned, reviewable, and re-importable into any watsonx Orchestrate or LangFlow environment — a single artifact judges can open and inspect directly.

09 Real-Time Match Advisor Workflow

Kratos supports the coaching staff across all three phases of a matchday.

BEFORE MATCH

Set the Plan

- Suggested formation & starting XI
 - Opponent scouting brief
 - Set-piece assignments
 - Confidence-scored risk notes
-

DURING MATCH

Adjust Live

- Substitution suggestions
 - Formation & pressing shifts
 - Momentum & fatigue flags
 - Live tactical alerts
-

AFTER MATCH

Close the Loop

- Performance vs. plan
- Key moment breakdown
- Player rating summary
- Lessons for the next cycle

10 Visualization Dashboard

A professional, IBM Hackathon–themed dashboard, responsive and available in dark mode throughout.

Pages

- Home
- Upload Match Data
- Opponent Analysis
- Formation Analysis
- Player Performance
- Heatmap
- Passing Network
- Tactical Board
- Strategy Recommendation
- Match Prediction
- Analytics Dashboard
- Coach Chatbot & Settings

Dashboard Features

- Interactive radar, bar, pie & line charts
- Possession & passing graphs
- Player rating cards
- Heat maps & formation diagrams
- Timeline & match comparison
- Opponent & player comparison
- Win probability
- AI recommendation cards

11 Kratos Conversational Assistant

The chatbot draws on the same RAG-grounded knowledge base as the dashboard, and always names which agent's analysis backs an answer.

Example interaction below — illustrative of the response structure Kratos returns, not a live scouting result.

Coach

How should we play against Barcelona?

Kratos

Built from the Opponent Strategy and Strategy Recommendation agents:

- **Recommended formation** — best-fit shape for this opponent
- **Key space to attack** — zone flagged by the Opponent Strategy Agent
- **Primary risk** — counter-attack exposure noted by Match Analysis
- **Confidence** — score, with the matches it's based on

Other things coaches ask Kratos: “*Why this formation?*” · “*Who should start up front?*” · “*Predict the score.*” · “*Where are we weak?*” · “*Best striker for this matchup?*”

12 Backend Architecture

A FastAPI backend organised around the four agents and the Kratos orchestrator.

Endpoint group	Purpose
/auth	Login & JWT token issuance
/agents/match-analysis	Match analysis queries
/agents/opponent-strategy	Opponent profiling
/agents/strategy	Tactical recommendations
/agents/insight	Plain-language explanations
/chat	Kratos conversational endpoint
/matches · /players · /teams	Knowledge-base CRUD & search

Cross-cutting concerns are handled once, centrally: JWT authentication, request validation, Swagger / OpenAPI documentation, structured logging, response caching, and rate limiting.

13 Frontend Architecture

ReactJS and Tailwind CSS, built for touchline tablets as much as the office desktop.

- Component-based architecture — reusable cards, charts and forms
- Tailwind CSS utility styling for a consistent, fast-to-iterate design system
- Responsive, tablet-first layouts for touchline use
- Dark mode throughout
- Loading and skeleton states for async agent calls
- Animated transitions for a polished, hackathon-ready feel

14 Security & Non-Functional Requirements

- JWT-based authentication & role checks
- Input validation on every endpoint
- Rate limiting against abuse
- Environment-variable configuration — no secrets in code
- Structured logging & centralised error handling
- Modular, SOLID-aligned codebase for maintainability

15Project Deliverables Checklist

This report is one deliverable within the full IBM Hackathon submission set defined by the problem statement.

Architecture diagram	✓ Included — Section 04
Multi-agent design & prompts overview	✓ Included — Section 05
RAG pipeline & knowledge base design	✓ Included — Section 06
IBM technology mapping	✓ Included — Sections 07–08
Project report / IBM submission report	✓ Included — this document
Future scope, limitations & conclusion	✓ Included — Sections 16–17
Complete source code — backend & frontend	Companion engineering deliverable
Docker setup	Companion engineering deliverable
README, Installation & Deployment guides	Companion engineering deliverable
Sequence, Flowchart, ER & Class diagrams	Companion technical annex
API documentation (Swagger)	Generated from the FastAPI backend
LangFlow JSON export	Companion engineering deliverable
PowerPoint deck & demo script	Companion deliverable

Items marked “Companion deliverable” are produced alongside the working codebase rather than inside a written report — ask if you’d like any of them built out next.

16Future Scope & Limitations

Future Scope

- Computer-vision ingestion of match video for automatic event tagging
- Wearable & GPS integration for live physical-load monitoring
- Expansion beyond football to other team sports
- IBM Cloud SaaS packaging for clubs & academies
- Native touchline mobile app
- Voice-driven, hands-free Kratos on matchday

Limitations

- Recommendation quality is bounded by the completeness of historical data
- Kratos is decision support, not a replacement for coaching judgment
- Live in-match suggestions are sensitive to latency
- New teams or players with little history are a cold-start challenge
- All tactical output should be reviewed by qualified coaching staff before use

17 Conclusion

Kratos shows what happens when Retrieval-Augmented Generation stops being a demo trick and becomes a working staff member.

Four specialist agents do narrow, well-defined jobs. One orchestrator agent gives the coaching staff a single, trustworthy voice. A full IBM stack — watsonx.ai, Granite, watsonx Orchestrate, and LangFlow — does the heavy lifting underneath. It is built to satisfy Problem Statement No. 29 end to end: analysis, opponent modelling, recommendation, and explanation, delivered through a dashboard and a conversational assistant a coach would actually want to open on a Tuesday morning before Saturday's match.